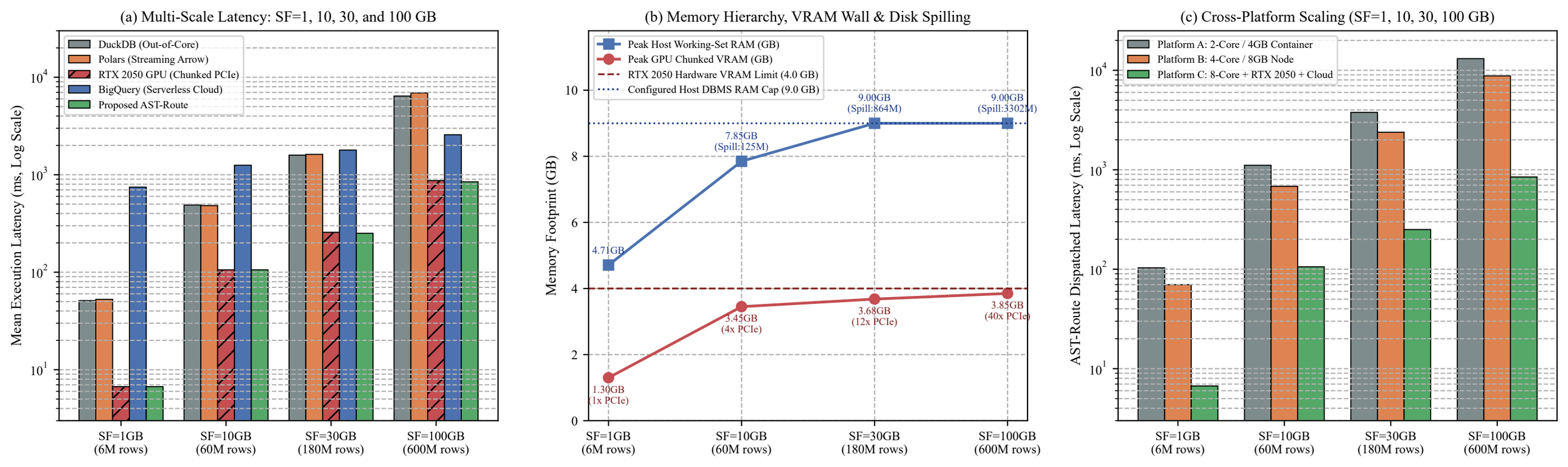
(a) Multi-Scale Latency: SF=1, 10, 30, and 100 GB
DuckDB (Out-of-Core)
Polars (Streaming Arrow)
RTX 2050 GPU (Chunked PCIe)
BigQuery (Serverless Cloud)
Proposed AST-Route
Mean Execution Latency (ms, Log Scale)
SF=1GB (6M rows)
SF=10GB (60M rows)
SF=30GB (180M rows)
SF=100GB (600M rows)
(b) Memory Hierarchy, VRAM Wall & Disk Spilling
Peak Host Working-Set RAM (GB)
Peak GPU Chunked VRAM (GB)
RTX 2050 Hardware VRAM Limit (4.0 GB)
Configured Host DBMS RAM Cap (9.0 GB)
Memory Footprint (GB)
4.71GB
7.85GB (Spill:125M)
9.00GB (Spill:864M)
9.00GB (Spill:3302M)
1.30GB (1x PCIe)
3.45GB (4x PCIe)
3.68GB (12x PCIe)
3.85GB (40x PCIe)
SF=1GB (6M rows)
SF=10GB (60M rows)
SF=30GB (180M rows)
SF=100GB (600M rows)
(c) Cross-Platform Scaling (SF=1, 10, 30, 100 GB)
Platform A: 2-Core / 4GB Container
Platform B: 4-Core / 8GB Node
Platform C: 8-Core + RTX 2050 + Cloud
AST-Route Dispatched Latency (ms, Log Scale)
SF=1GB (6M rows)
SF=10GB (60M rows)
SF=30GB (180M rows)
SF=100GB (600M rows)